Computer Engineering Department

Spring 2024 **EC372 Midterm Exam** **Time: 90 min**

Student Name:…… …….. Student ID: …….

- **Use 4 significant digits in your calculations.**

Q1) Use Taylor polynomial of the function $f(x)=\sqrt{x}$ from a = 2.0 to find *f*(2.5) for a relative true error of less than 0.05%. **(14 marks)**

$$f(a)+f'(a)(x-a)+\frac{f^{(2)}(a)}{2!}(x-a)^2+\frac{f^{(3)}(a)}{3!}(x-a)^3+\ldots$$

Q2)

False Position $x_m=x_u-\frac{f(x_u)(x_L-x_U)}{f(x_L)-f(x_U)}$	**Newton-Raphson** $x_{i+1}=x_i-\frac{f(x_i)}{f'(x_i)}$

a) Consider $f(x) = e^{-x}(3.2\sin(x) - 0.5\cos(x))$

*U*se the **false position method** to solve the equation on the interval [3, 4] (3 iterations). Calculate the relative approximate error at each iteration. **(18 marks)**

#	xl	f(xl)	Xu	f(xu)	Xm	f(Xm)	Relative approximation error (RAE)

b) For $f(x)=x^3-9x+2$Use the Newton-Raphson method to solve the equation (3 iterations, $x_0 = 0$). Calculate the relative approximate error at each iteration. **(18 marks)**

i	x_i	$f(x_i)$	$f'(x_i)$	x_{i+1}	$\lvert x_{i+1}-x_i\rvert$	RAE
0	0					

Q3) Solve the following linear system using LU decomposition **(25 marks)**

$x_1+x_2+x_3=1,$

$4x_1+3x_2-x_3=6,$

$3x_1+5x_2+3x_3=4$

L =	U=	Z=	X=

Computer Engineering Department

Spring 2024 **EC372 Midterm Exam** **Time: 90 min**

Student Name:…… ……... Student ID: …….

Q4) a) Use the linearized data model to find the constants of the regression model $y = ax\,e^{-bx}$ to best fit the following data set **(15 marks)**

x	1	2	3	4
y	2	3	4	4.5

b) Determine whether the curve a good fit for the data or not **(10 marks)**

$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$ $a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n}$	$S_t = \sum (y_i - \overline{y_{model}})^2$
$S_r = \sum_{i=1}^{n} (y_i - y_{model})^2$	$r = \sqrt{\frac{S_t - S_r}{S_t}}$